CS257 Linear and Convex Optimization
Homework 11

Due: November 30, 2020

December 7, 2020

1. Consider the following problem,

$$\begin{aligned}
\min_{\boldsymbol{x}\in\mathbb{R}^2} \quad & f(\boldsymbol{x}) = x_1^2 + x_2^2 \\
\text{s.t.} \quad & h_1(\boldsymbol{x}) = (x_1-1)^2 + (x_2-1)^2 - 1 = 0 \\
& h_2(\boldsymbol{x}) = (x_1-1)^2 + (x_2+\frac{1}{2})^2 - 1 = 0
\end{aligned}$$

(a). Write down the Lagrange condition.

(b). Find the points satisfying the Lagrange condition and the corresponding Lagrange multipliers.

(c). Which point is the optimal solution to the minimization problem? (Hint: A plot may be helpful.)

Solution. (a). The Lagrangian is

$$\mathcal{L}(x,\lambda) = x_1^2 + x_2^2 + \lambda_1[(x_1-1)^2 + (x_2-1)^2 - 1] + \lambda_2[(x_1-1)^2 + (x_2+\frac{1}{2})^2 - 1)]$$

The Lagrange condition is

$$\begin{cases}
\partial_{x_1}\mathcal{L} = 2x_1 + 2\lambda_1(x_1-1) + 2\lambda_2(x_1-1) = 0 \\
\partial_{x_2}\mathcal{L} = 2x_2 + 2\lambda_1(x_2-1) + 2\lambda_2(x_2+\frac{1}{2}) = 0 \\
\partial_{\lambda_1}\mathcal{L} = (x_1-1)^2 + (x_2-1)^2 - 1 = 0 \\
\partial_{\lambda_2}\mathcal{L} = (x_1-1)^2 + (x_2+\frac{1}{2})^2 - 1 = 0
\end{cases}$$

(b). Solutions to above equations

$$(1) = \begin{cases}
x_1 = 1 + \frac{\sqrt{7}}{4} \\
x_2 = \frac{1}{4} \\
\lambda_1 = -\frac{2}{\sqrt{7}} - \frac{1}{3} \\
\lambda_2 = -\frac{2}{\sqrt{7}} - \frac{2}{3}
\end{cases}
\qquad
(2) = \begin{cases}
x_1 = 1 - \frac{\sqrt{7}}{4} \\
x_2 = \frac{1}{4} \\
\lambda_1 = \frac{2}{\sqrt{7}} - \frac{1}{3} \\
\lambda_2 = \frac{2}{\sqrt{7}} - \frac{2}{3}
\end{cases}$$

(c). From Figure 1 we can see $(1-\frac{\sqrt{7}}{4}, \frac{1}{4})^T$ is the optimal solution and $(1+\frac{\sqrt{7}}{4}, \frac{1}{4})^T$ is the global maximum. We can also arrive at this conclusion by directly evaluating f at the two candidate solutions, since they

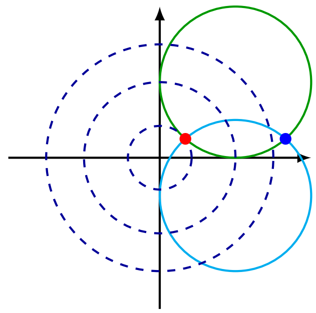

Figure 1

are the only two feasible points.

□

2. Consider the following problem that replaces the equality constraints in Problem 1 with inequality constraints,

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^2} \quad & f(\boldsymbol{x}) = x_1^2 + x_2^2 \\ \text{s.t.} \quad & g_1(\boldsymbol{x}) = (x_1-1)^2 + (x_2-1)^2 - 1 \le 0 \\ & g_2(\boldsymbol{x}) = (x_1-1)^2 + (x_2+\frac{1}{2})^2 - 1 \le 0 \end{aligned}$$

(a). Write down the KKT conditions.

(b). Do the optimal point you find in Problem 1(c) and its corresponding Lagrange multipliers satisfy the KKT conditions in part (a)? What can you conclude about the optimal solution to the inequality constrained problem?

Solution. (a). The KKT conditions are

$$\begin{cases} \mu_1 \ge 0 & \text{(1a)} \\ \mu_2 \ge 0 & \text{(1b)} \\ 2x_1 + 2\mu_1(x_1-1) + 2\mu_2(x_1-1) = 0 & \text{(1c)} \\ 2x_2 + 2\mu_1(x_2-1) + 2\mu_2(x_2+\frac{1}{2}) = 0 & \text{(1d)} \\ \mu_1[(x_1-1)^2 + (x_2-1)^2 - 1] = 0 & \text{(1e)} \\ \mu_2[(x_1-1)^2 + (x_2+\frac{1}{2})^2 - 1] = 0 & \text{(1f)} \\ (x_1-1)^2 + (x_2-1)^2 \le 1 & \text{(1g)} \\ (x_1-1)^2 + (x_2+\frac{1}{2})^2 \le 1 & \text{(1h)} \end{cases}$$

(b). The optimal solution and its Lagrange multipliers in Problem 1(c)

$$\begin{cases} x_1 = 1 - \frac{\sqrt{7}}{4} \\ x_2 = \frac{1}{4} \\ \mu_1 = \frac{2}{\sqrt{7}} - \frac{1}{3} \\ \mu_2 = \frac{2}{\sqrt{7}} - \frac{2}{3} \end{cases}$$

satisfies the KKT conditions. Condition (1a) and (1b) are obvious. The other conditions hold because x_1, x_2, μ_1, μ_2 are solutions to the Lagrange condition in Problem 1(a). Since the inequality constrained problem is convex, KKT conditions are sufficient for optimality, so $(1 - \frac{\sqrt{7}}{4}, \frac{1}{4})^T$ is the optimal solution for the inequality constrained problem as well. This is also clear from Figure 2.

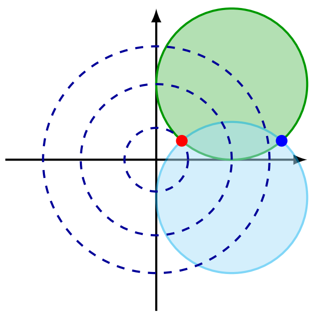

Figure 2

$\square$

3. Consider the following problem

$$\begin{aligned} \min \quad & (x_1 - \frac{9}{4})^2 + (x_2 - 2)^2 \\ \text{s.t.} \quad & -x_1^2 + x_2 \geq 0 \\ & x_1 + x_2 \leq 6 \\ & x_1, x_2 \geq 0 \end{aligned}$$

For each of the following points, determine whether it is an optimal solution to the above problem and show your arguments,

$$\boldsymbol{x}^{(1)} = \begin{bmatrix} \frac{9}{4} \\ 2 \end{bmatrix}, \boldsymbol{x}^{(2)} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}, \boldsymbol{x}^{(3)} = \begin{bmatrix} \frac{3}{2} \\ \frac{9}{4} \end{bmatrix}$$

(Hint: Try if you can find Lagrange multipliers satisfying the KKT conditions. Note you can easily check which constraints are active.)

Solution. The problem can be written as:

$$\begin{aligned} \min \quad & f(\boldsymbol{x}) = (x_1 - \frac{9}{4})^2 + (x_2 - 2)^2 \\ \text{s.t.} \quad & g_1(\boldsymbol{x}) = x_1^2 - x_2 \leq 0 \\ & g_2(\boldsymbol{x}) = x_1 + x_2 - 6 \leq 0 \end{aligned}$$

$$g_3(\boldsymbol{x}) = -x_1 \leq 0$$
$$g_4(\boldsymbol{x}) = -x_2 \leq 0$$

As both the objective function and the inequality constraints are convex, we only need to see if the given points satisfy the KKT condition.

- $\boldsymbol{x}^{(1)}$: $\boldsymbol{x}^{(1)}$ is not a feasible point, as

$$g_1(\boldsymbol{x}_1^{(1)}) = \frac{81}{16} - 2 > 0.$$

- $\boldsymbol{x}^{(2)}$: $\boldsymbol{x}^{(2)}$ is a feasible point and only $g_3(\boldsymbol{x}) \leq 0$ is active. The (reduced) KKT conditions are

$$\begin{cases} 2(x_1 - \frac{9}{4}) - \mu_3 = 0 \\ 2(x_2 - 2) = 0 \\ \mu_3 \geq 0 \end{cases}$$

  From the first equation we obtain $\mu_3 = 2(x_1^{(2)} - \frac{9}{4}) = -\frac{9}{2}$, violating $\mu_3 \geq 0$. There are no Lagrange multipliers satisfying the KKT conditions at $\boldsymbol{x}^{(2)}$. As $\nabla g_3(\boldsymbol{x}^{(2)}) = (-1, 0)^T \neq \mathbf{0}$, $\boldsymbol{x}^{(2)}$ is regular and the KKT conditions are necessary for optimality. Thus $\boldsymbol{x}^{(2)}$ cannot be optimal.

- $\boldsymbol{x}^{(3)}$: $\boldsymbol{x}^{(3)}$ is a feasible point and only $g_1(\boldsymbol{x})$ is active. The (reduced) KKT conditions are

$$\begin{cases} 2(x_1 - \frac{9}{4}) + 2\mu_1 x_1 = 0 \\ 2(x_2 - 2) - \mu_1 = 0 \\ \mu_1 \geq 0 \end{cases}$$

  which are satisfied by $\mu_1 = \frac{1}{2}$. Since the problem is convex and the KKT conditions hold, $\boldsymbol{x}^{(3)}$ must be optimal.

$\square$